A Google Earth Engine approach for **wildfire susceptibility prediction** fusion with remote sensing data of different spatial resolutions

S Tavakkoli Piralilou, G Einali, O Ghorbanzadeh… - Remote sensing, 2022 - mdpi.com

… (GPS)-based field **survey** points of **wildfire** locations. These locations … The modeling and **mapping** of **wildfire susceptibility** depend … influence the **wildfire** ignition and **spread** speed while …

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Mapping wildfire susceptibility with the BURN-P3 **simulation model**

MA Parisien, VG Kafka, KG Hirsch, JB Todd, SG Lavoie… - 2005 - researchgate.net

… Fire **Behavior Prediction** System and spatial data for forest fuels and topography with probabilistic fire ignitions and **spread** events derived from historical fire and weather data. **Model** …

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A **review** of planning and operational models used for emergency evacuation situations in Australia

MAP Taylor, SK Freeman - Procedia Engineering, 2010 - Elsevier

… represented on a **map**, and then these **maps** are overlaid to … of a **bushfire** risk **management model**. It is a scenario based … **model**; the CSIRO southern grassland fire **spread model** and …

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Mapping regional **forest fire** probability using artificial neural network **model** in a Mediterranean forest ecosystem

O Satir, S Berberoglu, C Donmez - Geomatics, Natural Hazards …, 2016 - Taylor & Francis

… This paper discusses **forest fire** probability **mapping** of a Mediterranean forestland using a … , back-**propagation** ANN described above is used as a regression **model** to estimate fire …

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Information systems in support of **wildland fire management** decision making in Canada

BS Lee, ME Alexander, BC Hawkes, TJ Lynham… - … and Electronics in …, 2002 - Elsevier

… of predicting **wildland fire** occurrence and **behavior** potential. … daily or hourly **maps** of fire weather, fire **behavior**, ignition … These analyses can be used to support the setting or **review** of …

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A Bayesian belief network analysis of factors influencing **wildfire** occurrence in Swaziland

WM Dlamini - Environmental Modelling & Software, 2010 - Elsevier

… **model** was developed from a set of 12 biotic, abiotic and human variables to determine factors that influence **wildfire** activity in Swaziland using **wildfire** … predictors of **wildfire** occurrence, …

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GIS-based evolutionary optimized Gradient Boosted Decision Trees for **forest fire susceptibility mapping**

S Sachdeva, T Bhatia, AK Verma - Natural Hazards, 2018 - Springer

… the **wildfire** locations recorded by the Forest **Survey** of India (available at http://fsi.nic.in/… prone land plays a crucial role in the progression of **wildfire** as fire tends to **spread** quicker, up the …

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Forest roads and operational **wildfire** response planning

MP Thompson, BM Gannon, MD Caggiano - Forests, 2021 - mdpi.com

… Supporting **wildfire management** activities is frequently … In this communication we briefly **review** and illustrate how … PCL **map** to a workshop along with other spatial fire **behavior** products …

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AEGIS: a **wildfire** prevention and **management** information system

K Kalabokidis, A Ager, M Finney… - … Hazards and Earth …, 2016 - nhess.copernicus.org

… The **wildfire** risk service generated hourly **maps** of the … , and to **model** the effect of fuel breaks on **wildfire spread**. Under … , AA: A **review** of recent advances in risk analysis for **wildfire** …

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A **review** of pathways for building fire **spread** in the wildland urban interface part II: response of components and systems and mitigation strategies in the United States

RSP Hakes, SE Caton, DJ Gorham, MJ Gollner - Fire technology, 2017 - Springer

… development in rural areas, changing fuel **management** policies, and climate change, … **spread** in the wildland urban interface. Part I discussed the fundamentals of **wildland fire spread** …

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Describing wildland surface fuel loading for fire **management**: a **review** of approaches, methods and systems

RE Keane - International journal of **wildland fire**, 2013 - connectsci.au

… **map** fuels for **wildland fire** research and **management**. As a result, many fire behaviour and effects software **prediction** … of fuel **dynamics** with the new input **model** requirements to allow …

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Wildfire susceptibility mapping: Deterministic vs. stochastic approaches

M Leuenberger, J Parente, M Tonini, MG Pereira… - … Modelling & Software, 2018 - Elsevier

Wildfire susceptibility is a measure of land propensity for the occurrence of wildfires based on terrain's intrinsic characteristics. In the present study, two stochastic approaches (ie, …

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Modeling the spatial variability of **forest fire susceptibility** using geographical information systems and the analytical hierarchy process

G Ljubomir, D Pamučar, S Drobnjak… - … in GIS and R for earth …, 2019 - Elsevier

… of this research was **forest fire susceptibility mapping** in the western part of the Republic of Serbia. A **forest fire** inventory **map** was developed using a serial historical **forest fire** database, …

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cffdrs: an R package for the Canadian **forest fire** danger rating system

X Wang, BM Wotton, AS Cantin, MA Parisien… - Ecological …, 2017 - Springer

… **wildland fire** ignition and **behavior** has been incorporated into the operational decision making within **wildland fire management** … We can then **map** a few key fire **behavior** measurements …

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Virtual Fire: A web-based GIS platform for **forest fire** control

K Kalabokidis, N Athanasis, F Gagliardi… - Ecological …, 2013 - Elsevier

… quantitative indices of **wildfire behavior** and effects with … of fire risk **maps** and **forecast** weather **maps**, which is applied … Virtual Fire provides the ability to monitor and **survey** several …

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A complex network theory approach for the spatial distribution of fire breaks in heterogeneous forest landscapes for the control of wildland fires

L Russo, P Russo, CI Siettos - PloS one, 2016 - journals.plos.org

… algorithm has also been used to simulate **wildland fire spread** [… **model** predicted quite well the **dynamics** of the large scale … and depict all simulations by **mapping** the burning frequency of …

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Wildland fire risk research in Canada

LM Johnston, X Wang, S Erni, SW Taylor… - Environmental …, 2020 - cdnsciencepub.com

… With this **review**, we aim to provide a better understanding of … influences on fire ignition and **spread**, subjective perceptions of … **simulation model** that produces burn probability **maps** (but …

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Defining **wildfire susceptibility maps** in Italy for understanding seasonal **wildfire** regimes at the national level

A Trucchia, G Meschi, P Fiorucci, A Gollini, D Negro - Fire, 2022 - mdpi.com

… Another approach involves the iteration of **wildfire spread model** runs with random ignition points and weather conditions, see, eg, [14,15]. Recent advances in machine learning (ML) …

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Characterizing potential **wildland fire** fuel in live vegetation in the Mediterranean region

S Fares, S Bajocco, L Salvati, N Camarretta… - Annals of Forest …, 2017 - Springer

… **Wildland fire management** is subject to manifold sources of uncertainty. Beyond the difficulties of predicting accurately the fire **behavior**… , classifying, and **mapping** fuel based on remotely …

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Negative consequences of positive feedbacks in US **wildfire management**

DE Calkin, MP Thompson, MA Finney - Forest Ecosystems, 2015 - Springer

… , infrared **mapping**, satellite detection, computer dispatching, fire … article we **review** the structure of US **wildfire management** … the **trajectory** of **wildfire** risk and associated **management** …

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